Can You Use Theorem 9 When γ^β Is in the Middle? Yes — Here Is Why

PHYS 7502

1 The Concern

In the interference trace, we need to evaluate:

$$\gamma^\alpha \, (\not{p}_1 + \not{p}_2) \, \gamma^\beta \, \not{p}_1 \, \gamma_\alpha \tag{1}$$

I claimed this is Theorem 9: $\gamma^\mu \not{a} \not{b} \not{c} \gamma_\mu = -2\not{c}\not{b}\not{a}$.

But γ^β is a **bare gamma matrix with a free index** — it is *not* a slashed momentum $p_\mu \gamma^\mu$. So how can we treat it as the "$\not{b}$" in the theorem?

2 The Answer: The Theorem Is Really About Bare Gammas

Theorem 9: The Real Version (Griffiths §7.7)

The fundamental form of Theorem 9 is stated in terms of **bare gamma matrices**:

$$\boxed{\gamma_\mu \, \gamma^\nu \, \gamma^\lambda \, \gamma^\sigma \, \gamma^\mu = -2 \, \gamma^\sigma \, \gamma^\lambda \, \gamma^\nu} \tag{2}$$

This works for **any** indices ν, λ, σ — free or contracted, it does not matter. The only requirement is that μ is the contracted pair (one upper, one lower, summed over).

The "slashed" version $\gamma^\mu \not{a} \not{b} \not{c} \gamma_\mu = -2\not{c}\not{b}\not{a}$ is a **consequence**, obtained by contracting ν, λ, σ with three 4-vectors a^ν, b^λ, c^σ. But the bare-gamma version (2) is more general.

The Key Point

γ^β **is** a gamma matrix. It is exactly the kind of object that appears in (2). The fact that β is a free index rather than being contracted with a momentum vector is irrelevant — Theorem 9 does not care what is happening to the indices ν, λ, σ. It only cares that μ is contracted.

A slashed momentum $\not{p} = p_\nu \gamma^\nu$ is just a *linear combination* of bare gammas (weighted by the components of p). The theorem works for each bare gamma individually, so by

linearity it works for any linear combination.

3 Proving It Explicitly for Our Case

Let us expand everything in (1) into bare gammas and apply Theorem 9 term by term.

3.1 Expand the slashed momenta

$$\not{p}_1 + \not{p}_2 = p_{1\nu}\gamma^\nu + p_{2\rho}\gamma^\rho \tag{3}$$

$$\not{p}_1 = p_{1\sigma}\gamma^\sigma \tag{4}$$

(Here ν, ρ, σ are dummy indices, each internally summed.)

3.2 Substitute into the expression

$$\gamma^\alpha \left(\not{p}_1 + \not{p}_2\right) \gamma^\beta \not{p}_1 \, \gamma_\alpha = \gamma^\alpha \left(p_{1\nu}\gamma^\nu + p_{2\rho}\gamma^\rho\right) \gamma^\beta \left(p_{1\sigma}\gamma^\sigma\right) \gamma_\alpha \tag{5}$$

Pull the momentum components (which are just numbers) out front:

$$= p_{1\nu}p_{1\sigma}\, \gamma^\alpha\gamma^\nu\gamma^\beta\gamma^\sigma\gamma_\alpha \;+\; p_{2\rho}p_{1\sigma}\, \gamma^\alpha\gamma^\rho\gamma^\beta\gamma^\sigma\gamma_\alpha \tag{6}$$

3.3 Apply the bare-gamma Theorem 9 to each term

Each term has the form $\gamma^\alpha\gamma^A\gamma^B\gamma^C\gamma_\alpha$ with α contracted. By (2):

$$\gamma^\alpha \, \gamma^A \, \gamma^B \, \gamma^C \, \gamma_\alpha = -2 \, \gamma^C \, \gamma^B \, \gamma^A \tag{7}$$

First term ($A = \nu$, $B = \beta$, $C = \sigma$):

$$\gamma^\alpha\gamma^\nu\gamma^\beta\gamma^\sigma\gamma_\alpha = -2\, \gamma^\sigma\gamma^\beta\gamma^\nu \tag{8}$$

Second term ($A = \rho$, $B = \beta$, $C = \sigma$):

$$\gamma^\alpha\gamma^\rho\gamma^\beta\gamma^\sigma\gamma_\alpha = -2\, \gamma^\sigma\gamma^\beta\gamma^\rho \tag{9}$$

3.4 Put the momentum components back

Substituting back into (6):

$$\begin{aligned}
&= p_{1\nu}p_{1\sigma}\, (-2\gamma^\sigma\gamma^\beta\gamma^\nu) + p_{2\rho}p_{1\sigma}\, (-2\gamma^\sigma\gamma^\beta\gamma^\rho) \\
&= -2 \, \underbrace{p_{1\sigma}\gamma^\sigma}_{\not{p}_1} \, \gamma^\beta \, \underbrace{\left(p_{1\nu}\gamma^\nu + p_{2\rho}\gamma^\rho\right)}_{\not{p}_1+\not{p}_2} \\
&= -2\, \not{p}_1 \, \gamma^\beta \left(\not{p}_1 + \not{p}_2\right)
\end{aligned} \tag{10}$$

Result

$$\boxed{\gamma^{\alpha}\,(\not{p}_1+\not{p}_2)\,\gamma^{\beta}\,\not{p}_1\,\gamma_{\alpha}=-2\,\not{p}_1\,\gamma^{\beta}\,(\not{p}_1+\not{p}_2)} \tag{11}$$

This is exactly Theorem 9 with the three middle objects reversed:

$$(\not{p}_1+\not{p}_2),\ \gamma^{\beta},\ \not{p}_1 \quad\longrightarrow\quad \not{p}_1,\ \gamma^{\beta},\ (\not{p}_1+\not{p}_2)$$

The bare γ^{β} (with its free index) participated in the reversal just like any other gamma matrix.

4 Why It Works: The Deeper Reason

All the Contraction Theorems Work on Bare Gammas

The contraction theorems are **identities about products of gamma matrices**. They are derived purely from the anticommutation relation $\{\gamma^{\mu},\gamma^{\nu}\}=2g^{\mu\nu}$ and the fact that μ runs over 4 values. They do not care whether the other indices are:

- **Free** (like γ^{β}) — the theorem treats β as a label; the identity holds for each value of β separately.
- **Contracted with a momentum** (like $\not{p}=p_{\nu}\gamma^{\nu}$) — by linearity, the identity holds for any linear combination.
- **Contracted with another gamma somewhere else in the trace** — the theorem is local to the $\gamma^{\mu}\cdots\gamma_{\mu}$ pair and does not touch indices elsewhere.

Slashed notation is a convenience, not a requirement. Writing $\not{a}\not{b}\not{c}$ just means "three gamma matrices, each contracted with a 4-vector." But the theorems work for any three gamma matrices in the middle, regardless of what their indices are doing.

5 Summary

Bottom Line

Q: Can you use Theorem 9 when γ^{β} (a bare gamma with a free index) sits in the middle?
A: Yes. Theorem 9 in its fundamental form is:

$$\gamma_{\mu}\gamma^{\nu}\gamma^{\lambda}\gamma^{\sigma}\gamma^{\mu}=-2\gamma^{\sigma}\gamma^{\lambda}\gamma^{\nu}$$

It works for **any** γ-matrices between the contracted μ pair — slashed momenta, bare gammas, or any mix. The "slashed" version is just a special case obtained by contracting with 4-vectors.
γ^{β} is a perfectly valid gamma matrix. It just happens to carry a free index that will be contracted *later* (by the photon polarization sum). Theorem 9 does not care about that

— it only contracts the α pair and reverses everything between.